

Supporting Information for: “Climate change impacts on compound renewable energy droughts under evolving infrastructure in the Western United States“

C. Bracken¹, N. Voisin^{1,2}, K. Mongird¹, C. D. Burleyson¹, K. Oikonomou¹

¹Pacific Northwest National Laboratory, Richland, WA USA

²University of Washington, Seattle, WA USA

Business-as-Usual (BAU) Scenario Results

This supporting information provides a table figures analagous to those in the paper but for the business-as-usual scenario. Please refer to the main paper for details on the differences between the scenarios.

Table of Contents

Table S1 – Installed capacities for the BAU scenario.

Figure S1-S3 – Energy drought statistics for the business-as-usual scenario.

Figure S4 – Energy drought connectivity across BAs for the business-as-usual scenario.

Figure S5 – The change in energy drought connectivity for the business-as-usual scenario.

BA	BA Name	solar 2020	solar 2035	solar 2050	wind 2020	wind 2035	wind 2050
AVA	Avista	0.0	0.2	1.1	0.2	1.8	1.8
AZPS	Arizona Public Service Company	0.8	4.1	4.4	0.2	4.0	8.3
BPAT	Bonneville Power Administration	0.1	1.6	1.5	3.4	8.4	6.6
CISO	California Independent System Operator	14.8	40.4	42.6	5.8	27.9	31.5
IPCO	Idaho Power Company	0.3	2.3	5.2	0.7	4.1	5.3
LDWP	L.A. Department of Water and Power	1.0	2.2	2.0	0.4	4.8	5.6
NEVP	Nevada Power Company	1.6	14.1	18.7	0.1	0.6	0.7
NWMT	NorthWestern Energy	0.0	6.9	10.6	0.5	18.0	19.9
PACE	PacifiCorp East	1.3	42.2	58.7	2.7	18.7	26.1
PACW	PacifiCorp West	0.3	2.0	1.7	0.7	1.3	0.9
PGE	Portland General Electric	0.1	0.2	0.1	0.7	1.0	0.6
PNM	Public Service Company of New Mexico	0.4	6.1	7.6	1.1	3.0	4.7
PSCO	Public Service Company of Colorado	0.5	10.7	20.2	4.5	11.2	18.1
PSEI	Puget Sound Energy	0.0	0.1	0.3	0.5	2.1	2.1
SRP	Salt River Project	0.3	1.2	1.1	0.1	0.1	0.1
TEPC	Tuscon Electric Power Company	0.3	4.7	8.3	0.0	0.4	0.9
WACM	WAPA - Colorado-Missouri	0.2	6.9	12.1	0.8	7.8	7.0
WALC	WAPA - Lower Colorado	0.1	2.8	6.7	0.3	3.8	4.1

Table S1. Balancing authorities used in the business-as-usual scenario in this study, with sited wind and solar capacity in gigawatts. *Western Area Power Administration

Corresponding author: Cameron Bracken, cameron.bracken@pnnl.gov

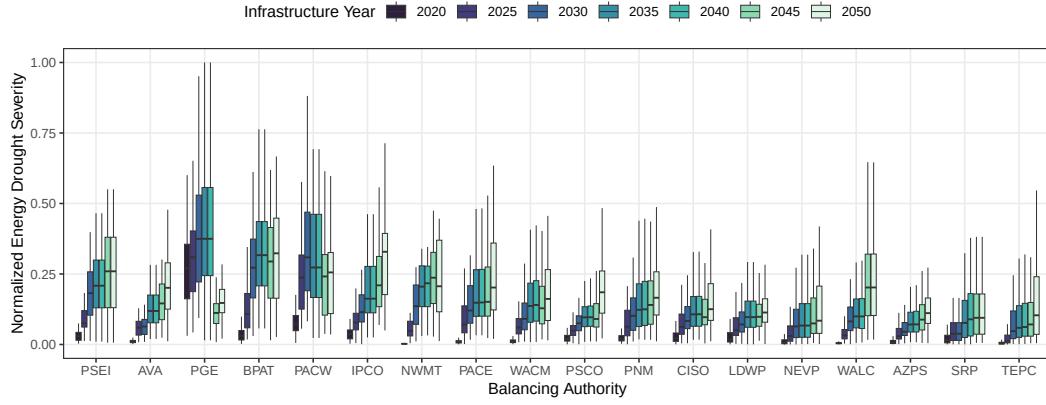


Figure S1. Energy drought severity (energy deficit below the 10th percentile) for the business-as-usual scenario which involves a fully decarbonized grid by 2050. Climate variability is simulated for each infrastructure year using 40-years of rcp85hotter climate forcing. Severity has been normalized by the maximum value in each BA to allow the data to be compared across BAs and outliers have been removed for visual clarity.

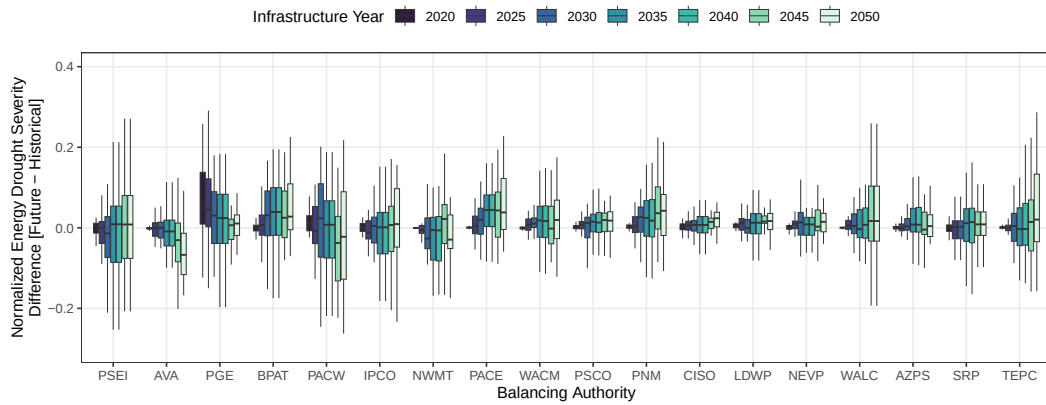


Figure S2. Difference in average annual normalized energy drought severity between the future and historical periods for the business-as-usual scenario, which involves a fully decarbonized grid by 2050. Severity has been normalized per BA to allow the data to be compared across BAs.

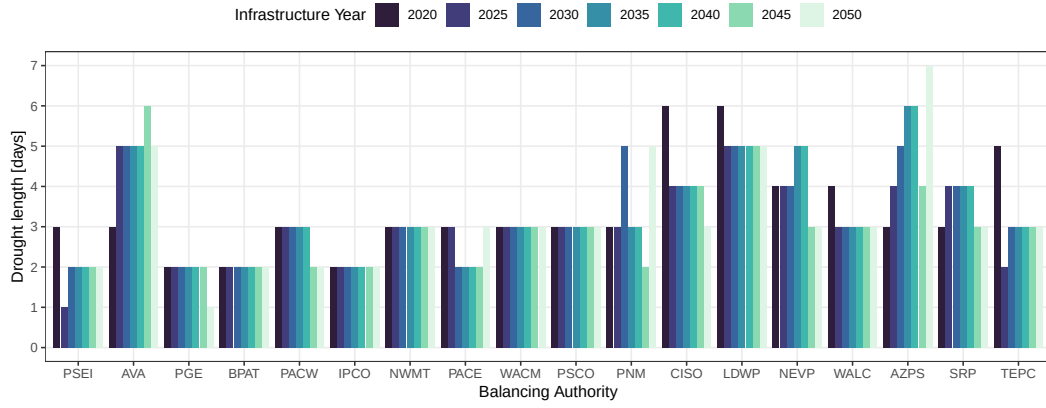


Figure S3. Energy drought duration for the business-as-usual scenario, which involves a fully decarbonized grid by 2050. Note that the figure does show boxplots, but in most BAs the distribution of drought duration is heavily skewed such that most boxes are collapsed on at the lowest value.

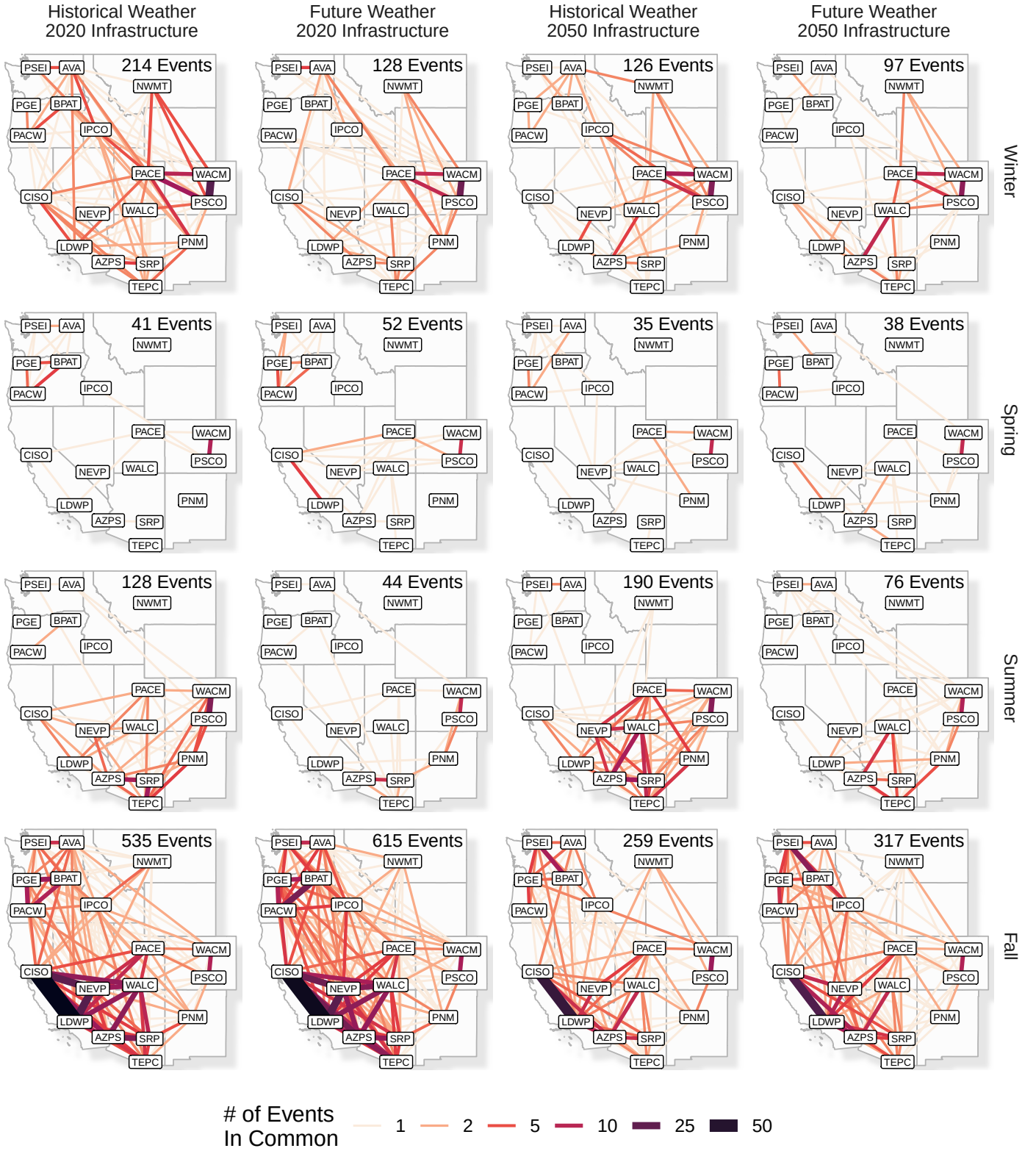


Figure S4. Seasonal compound energy drought co-occurrence between BAs for the business-as-usual scenario using historical weather with 2020 infrastructure (first column) and future weather with 2020 infrastructure (second column), historical weather with 2050 infrastructure (third column), and future weather with 2050 infrastructure (fourth column). The rows indicate the seasons. A line is drawn between two BAs if at least one energy drought occurred on the same day. The thickness and color of the line represent the number of events in common between a pair of BAs.

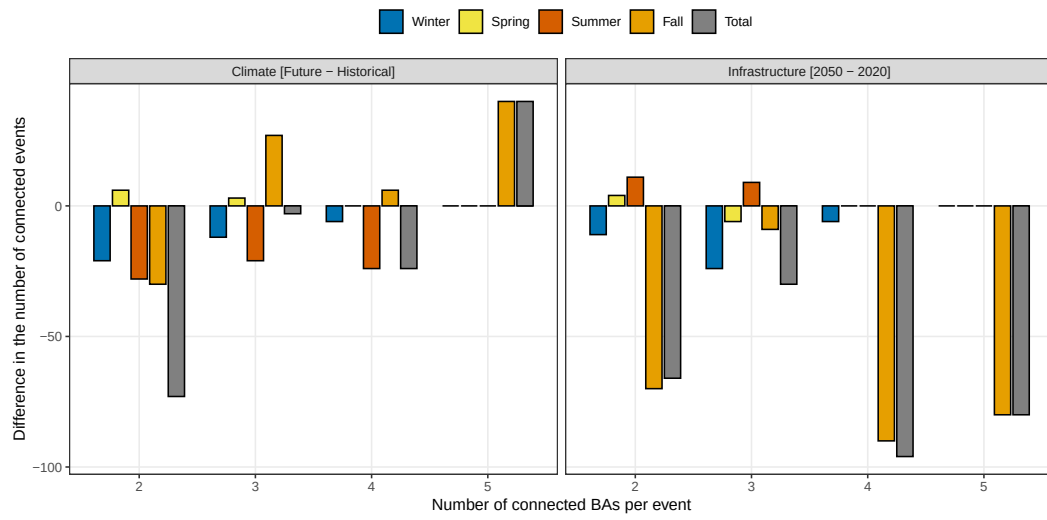


Figure S5. The difference between the number of multi-BA compound energy droughts in each season for the business-as-usual scenario. The left panel shows the climate influence by differencing the future weather and historical weather periods using 2050 infrastructure. The right panel shows the infrastructure influence by differencing the 2050 and 2020 infrastructure scenarios both using future weather. Negative values indicate fewer connected events in the future period.